



INTER IIT TECH MEET 14.0

Autonomous Mobile Robot for Warehouse Inventory

Team ID- 42



Problem & Motivation: Why AMRs?



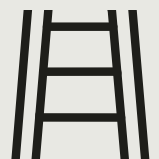
Low Productivity

Manual tasks lead to slow operations and high cycle count times.



Human Errors

Fatigue and variable scanning cause inaccurate inventory records.



Limited Accessibility

Inefficient manual shelf access

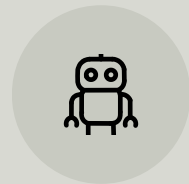


Scalability Issues

Manual workflows don't scale with growing SKU volumes.

Executive Summary: Revolutionizing Inventory

The solution aims to design an Autonomous Mobile Robot (AMR) for warehouse navigation and vertical scanning of storage racks up to 1.8 meters high. Equipped with a high-resolution visual data capture system, the AMR processes and stores information in a real-time database for inventory analysis.



Automated Tasks

AMRs handle repetitive and physically demanding tasks, reducing accidents.



Increased Accuracy

Enhanced precision in inventory tracking and data collection.



Faster Operations

Rapid cycle counts and continuous updates to inventory databases.



Solution Approach & Key Metrics

Our AMR integrates SLAM-based navigation, QR scanning, a vertical lift, and data logging. Built with off-the-shelf components, it combines differential drive motion, a camera-based scanning module, and ROS 2 powered algorithms.

Positioning Accuracy (X-Y)

±25 cm over 10 meters.

Vertical Lift Accuracy (Z)

Less than 2 cm precision with stepper motor.

QR Scanning Speed

Less than 1 second per QR code.

Full Rack Scan

Within 3 minutes, including motion and positioning.

Control Flow: High & Low Level

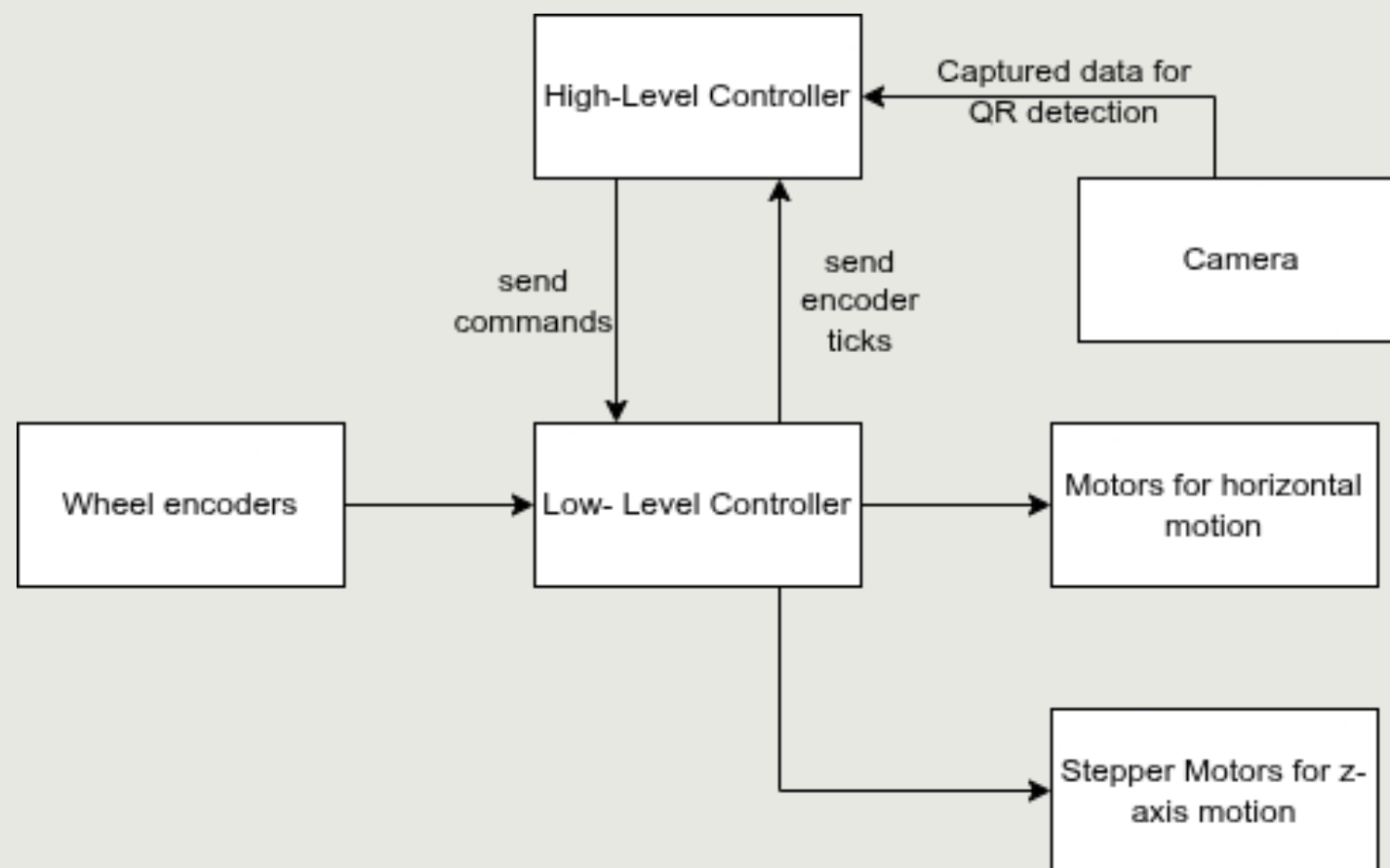
Low-Level Controller (Arduino Mega)

Provides real-time performance with sufficient GPIO pins and strong interrupt handling. Handles PWM signals and wheel encoder feedback.

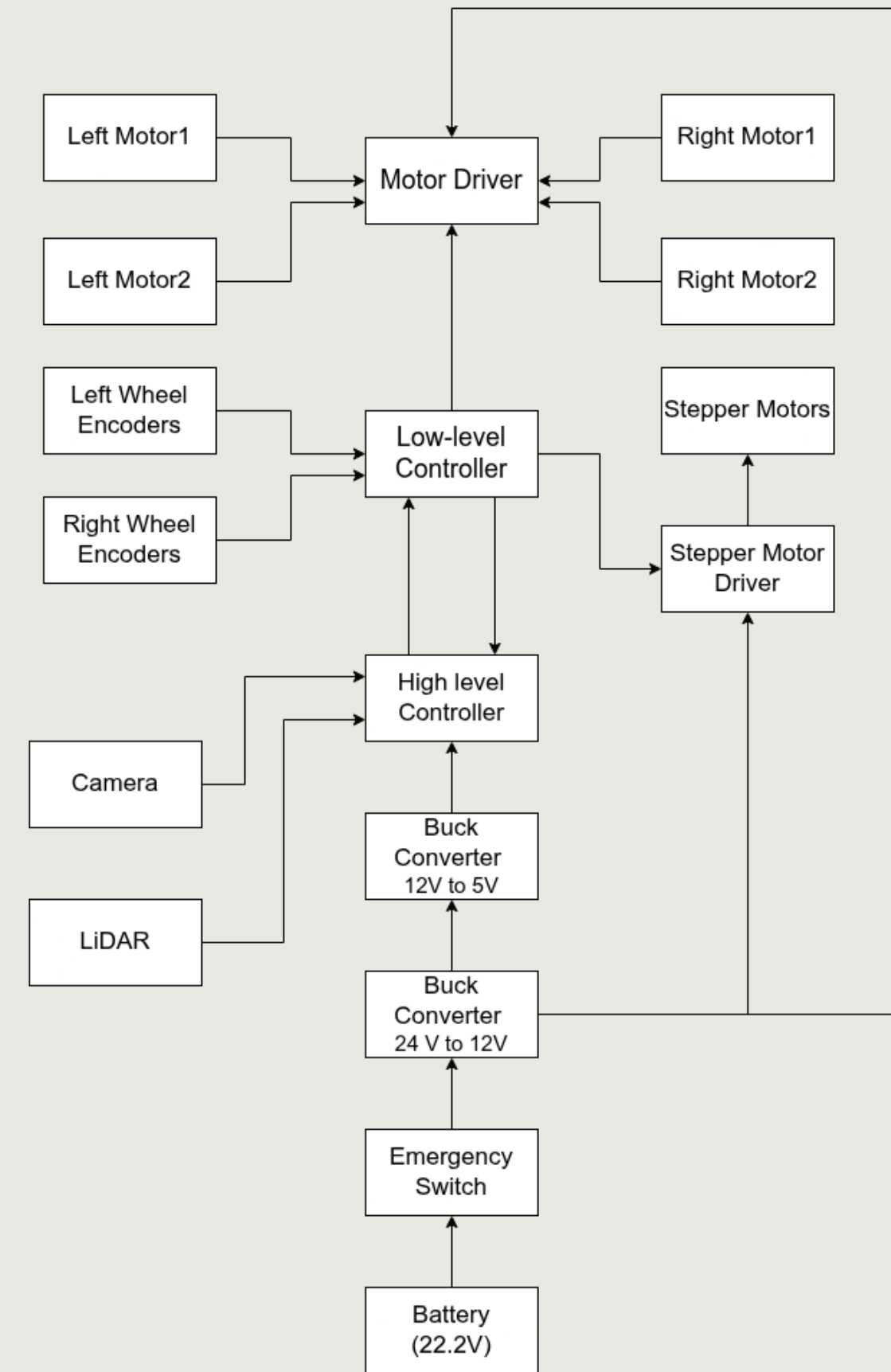
High-Level Controller

Functions as the computational unit for autonomous navigation, perception, and data processing. Runs ROS 2 middleware for SLAM, localization, and path planning.

The High-Level Controller processes LiDAR data for mapping and visual data for QR code detection, storing information in warehouse servers.



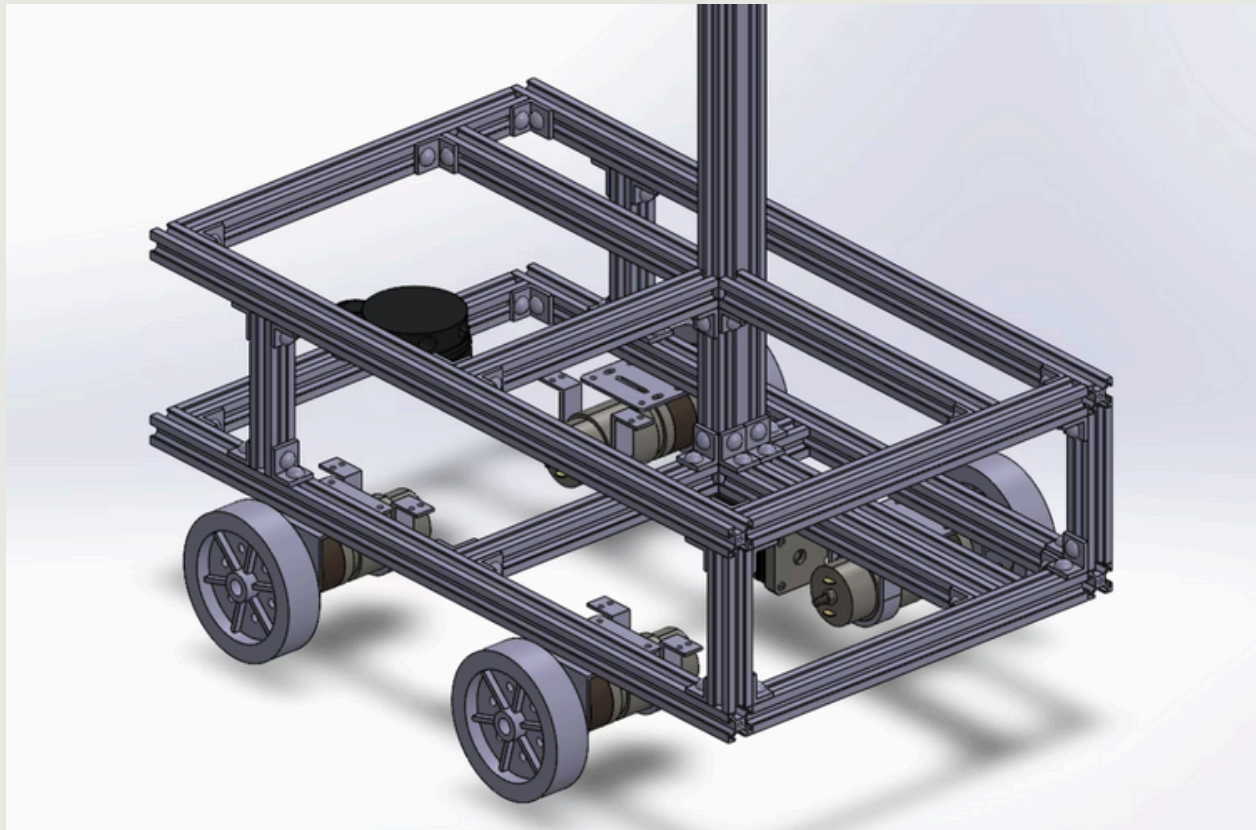
Inter-connections between the various components used



Hardware Design: Motion Systems

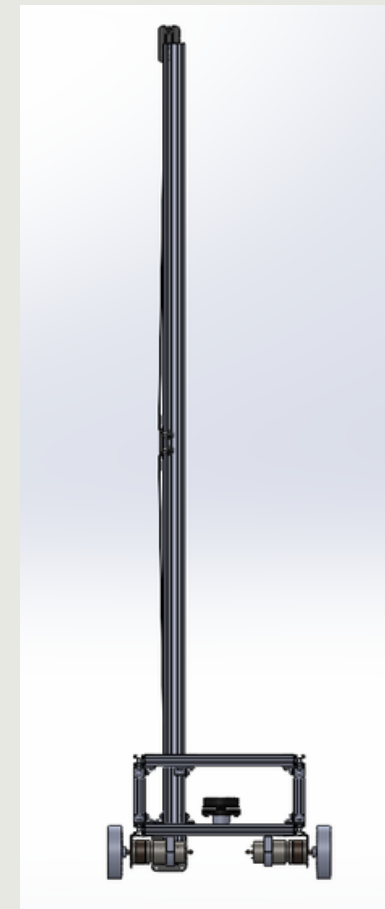
Horizontal Motion: Differential Drive

Four DC motors provide high maneuverability and precise control. Independent control allows forward/backward movement, turns, and zero-radius pivots for navigating narrow aisles.



Vertical Motion: Belt-Driven Lift

A 1.5-meter aluminum double-extrusion profile guides a belt-driven linear actuation system. A NEMA stepper motor drives a timing belt, moving a V-slot gantry plate with the camera module.



Lift Mechanism: A Closer Look

How does it work?

- A stepper motor drives a timing pulley connected to a reinforced belt running along the extrusion.
- An idler pulley at the top maintains belt tension and eliminates backlash.
- The belt is fixed to the V-slot gantry plate, so motor rotation converts directly into vertical motion.
- The gantry plate uses four precision rollers to ensure stable, wobble-free travel along the extrusion.
- This setup allows the camera module to move smoothly across the full 1.6-meter vertical range for scanning.



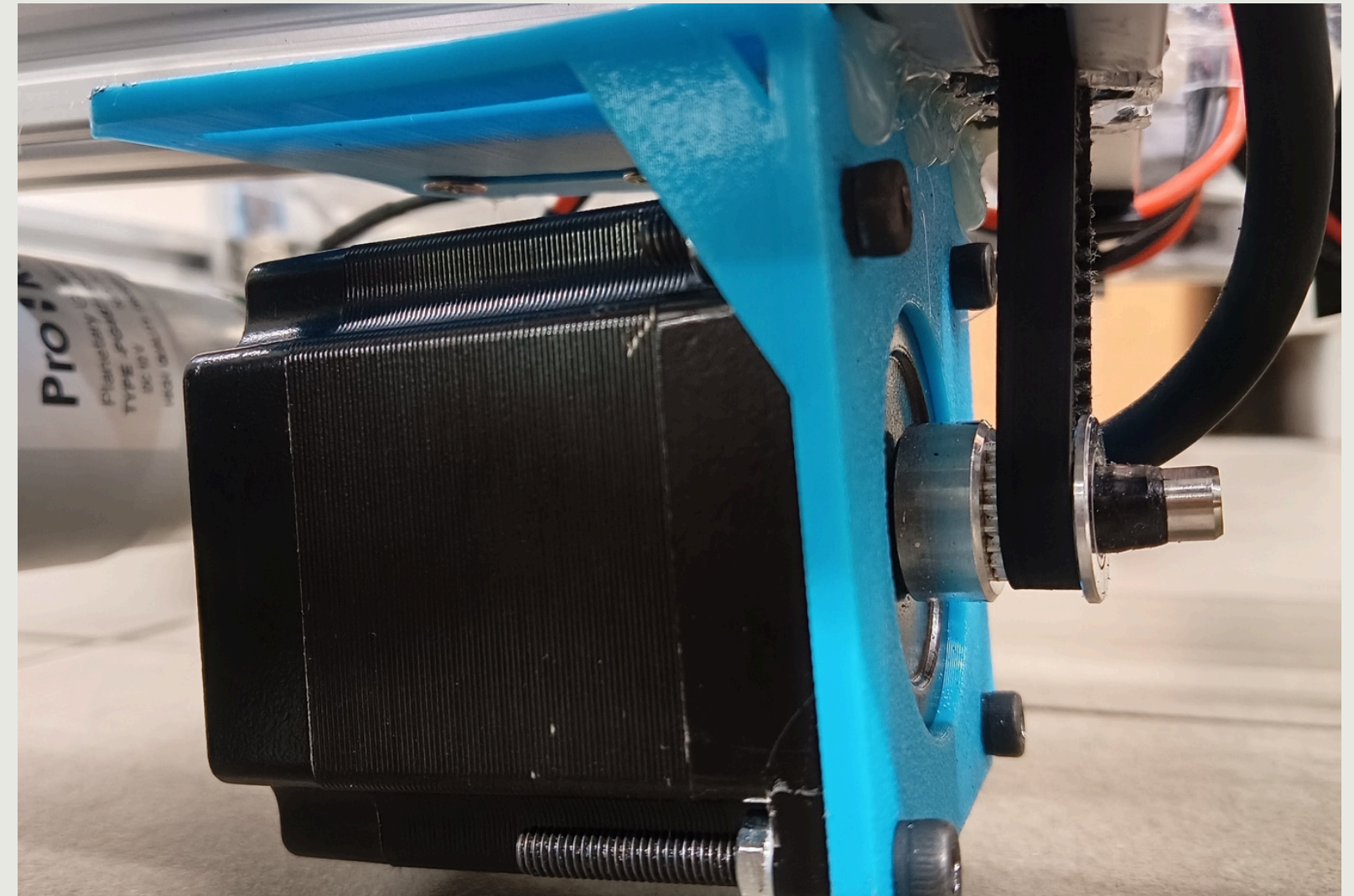
Pulley & Idler plate



V-slot gantry plate

Why use belt-driven lift?

- Provides high-speed, low-noise, and low-vibration motion suitable for precise camera scanning.
- Avoids lead-screw limitations such as screw whip, high friction, reduced speed over long travel, and increased torque demand for tall lifts.
- More efficient than telescopic masts, offering continuous, backlash-free motion without complex multi-stage assemblies.
- Enables lighter construction, easier maintenance, and better cost-performance for long-range vertical actuation (up to ~1.8 meters).



Stepper with pulley controlling linear actuation of belt

Gallery



Software Architecture

The AMR's software is built on a modular, ROS 2-based architecture, integrating perception, motion control and navigation. The high level controller handles high-level tasks like localization, path planning etc. while the Arduino Mega manages low-level actuation and sensing.

01

ROS 2 Operating Framework

Real-time communication, sensor integration, navigation algorithms, and modular scalability.

03

Navigation Stack (Nav2)

Handles global path planning, local obstacle avoidance, and collision-free trajectory computation.

02

SLAM & Localization

Uses slam_toolbox for 2D LiDAR-based SLAM, pose-graph optimization, and real-time map correction. AMCL localization for localizing the robot

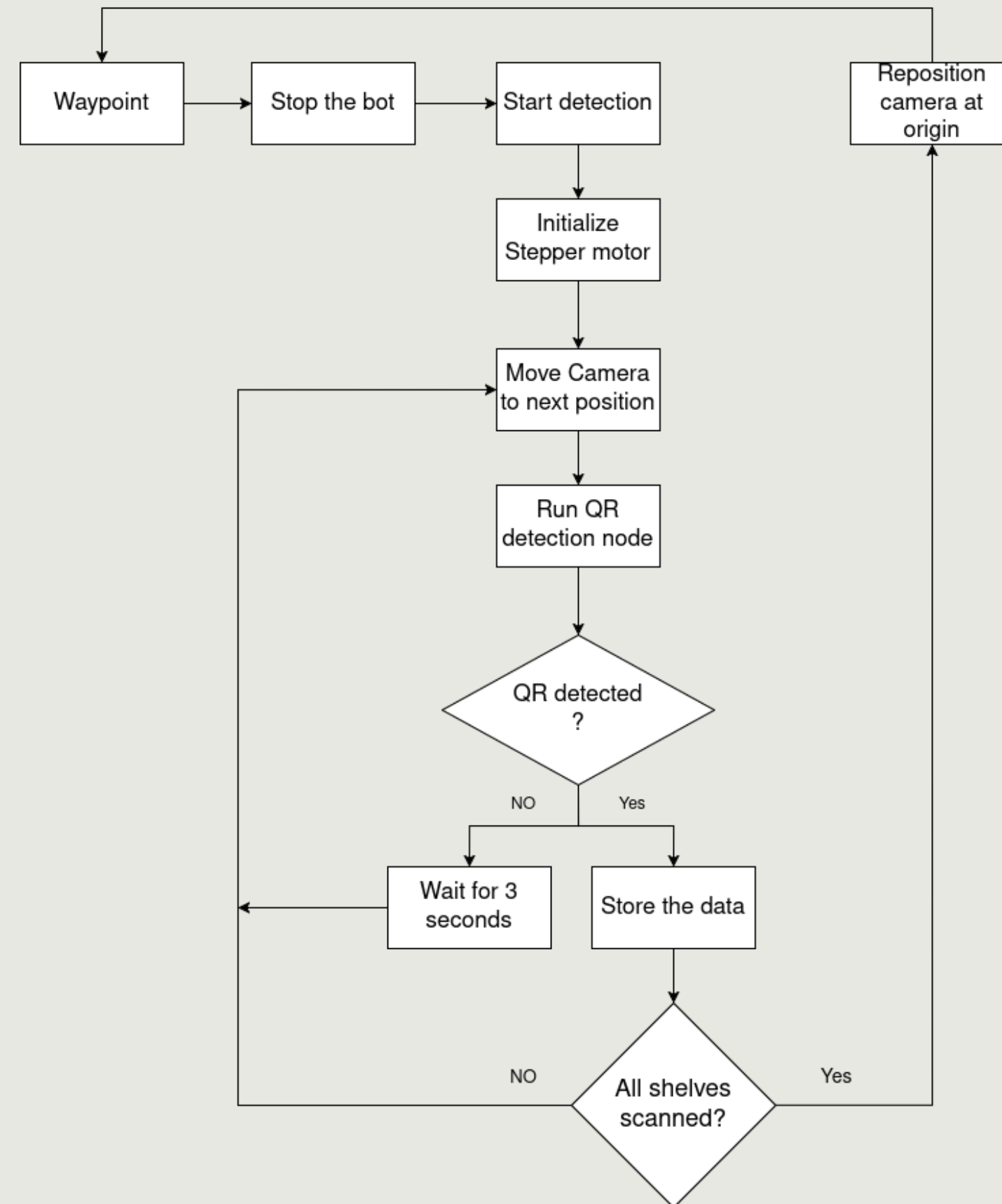
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Vertical Actuation & QR Scanning

Stepper-driven Z-axis lift positions camera at predefined shelf heights. OpenCV + Python script detects and decodes QR codes and stores information locally.

QR code scanning and Camera Control mechanism

On reaching the racks, the following processes are being followed for detection of QR codes and acquiring the data stored in them and managing the inventory.





Testing & Validation

Structured testing in a simulated warehouse environment validated the AMR's performance, reliability, and robustness across various parameters.

1

Motion & Navigation

Positional accuracy of ± 25 cm over 10m traversals. Max speed 0.6 m/s, stopping in 2 seconds.

2

Vertical Lift System

Less than 2 cm vertical positioning accuracy, 10 mm/s lift speed for quick transitions.

3

Perception & QR Scanning

QR code detection under 1 second. Full rack scan within 3 minutes.

4

System Control Stability

Motion control loop operates at 20 Hz for stable autonomous behavior.

5

Power & Endurance

Up to 1 hour of continuous operation under extreme condition (when using 22.2v ,20Ah battery).

Future Improvements

- Industrial level chasis building.
- Recovery Behaviour.
- Advanced scanning: dynamic shelf-height detection, adaptive camera, visual SLAM fusion.
- Implementing Autonomous Docking.

THANK YOU